

■ Alumni Web Application — Project Documentation

1. Project Overview

The Alumni Portal is a web-based application that connects past students, current professionals, and institutions (colleges, schools, or offices). It allows alumni to register, share career updates, and stay connected. Admins can manage all alumni data, while users (alumni) can view, search, and contact fellow alumni easily.

2. Objectives

- Build a unified platform for alumni networking.
- Allow institutions to manage and showcase alumni achievements.
- Enable secure admin and user logins.
- Provide search and filtering options for finding alumni by profession, batch, or organization.
- Simplify data collection via public forms linked to institutions.

3. Technology Stack

Frontend: React.js, Tailwind CSS / Bootstrap

Backend: Node.js with Express.js

Database: MongoDB / MySQL

Authentication: JWT or Session-based login

Hosting: Vercel / Render / Railway

Image Upload: Multer (Node.js)

Communication: Email or Contact Form integration

4. Modules

Admin Module: Manage alumni data, CRUD operations, approve new submissions.

Alumni Module: Login, view, and search alumni, contact features.

Public Module: Display alumni details, submission form linked to institutions.

5. Functionalities

- Admin Login, Add/Edit/Delete Alumni
- Search and Filter Alumni
- Dynamic Home Page Display
- Contact Option
- Password Hashing and Security
- Responsive UI

6. Database Design (Example)

alumni: name, email, profession, company, college, department, location, status, photo

users: username, email, password, role

7. Data Flow

1. User or Alumni fills form
2. Data sent to backend
3. Admin reviews and approves
4. Approved entries show on homepage
5. Alumni can log in and search others

8. Security Features

- Encrypted passwords (bcrypt)
- Protected routes
- Validation & Sanitization

- Admin-only modification access

9. Future Enhancements

- Add chat or message feature
- Integrate LinkedIn API
- Add events and job posting sections
- Enable real-time notifications
- Mobile app version

10. Conclusion

The Alumni Web App creates a strong digital bridge between institutions and their alumni, making it easier to stay connected, showcase achievements, and build professional relationships. Its modular design allows easy customization for any college, school, or organization.